

The corresponding equations are: 1. $7x_1 + 8x_4 = -11$ 2. $7x_2 - 2x_4 = -6$

3. $7x_3 - x_4 = 4$

Column 4 is not a pivot column. $\therefore x_4$ is free, by defn. Let $x_4 = -4$

$$\therefore 7x_1 + 8(-4) = -11 \Rightarrow 7x_1 = 21 \Rightarrow \boxed{x_1 = 3}$$

$$\therefore 7x_2 - 2(-4) = -6 \Rightarrow 7x_2 = -14 \Rightarrow \boxed{x_2 = -2}$$

$$\therefore 7x_3 - (-4) = 4 \Rightarrow 7x_3 = 0 \Rightarrow \boxed{x_3 = 0}$$

So, $p = \begin{bmatrix} 3 \\ -2 \\ 0 \\ -4 \end{bmatrix}$ is a solution of $Ax = b$.

$$\begin{bmatrix} 1 & 5 & -2 & 0 \\ -3 & 1 & 9 & -5 \\ 4 & -8 & -1 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} -7 \\ 9 \\ 0 \end{bmatrix}$$

By definition of Ax : $3 \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} + (-2) \begin{bmatrix} 5 \\ 1 \\ -8 \end{bmatrix} + 0 \begin{bmatrix} -2 \\ 9 \\ -1 \end{bmatrix} + (-4) \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$

$$= \begin{bmatrix} -7 \\ 9 \\ 0 \end{bmatrix}$$

2. Let $A = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$, $u = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$, and $v = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$. Verify Theorem 5(a) in this case by computing $A(u+v)$ and $Au + Av$.

Solution: $u+v = \begin{bmatrix} 4+(-3) \\ (-1)+5 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ (Vector addition)

$$A(u+v) = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 4 \\ 3 \cdot 1 + 1 \cdot 4 \end{bmatrix} \quad \begin{matrix} \text{(Row vector rule for computing } Ax) \end{matrix}$$

$$= \begin{bmatrix} 22 \\ 7 \end{bmatrix} \quad Au = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (-1) \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad \begin{matrix} \text{(By Defn. of } Ax) \end{matrix}$$

$$= \begin{bmatrix} 4 \cdot 2 \\ 4 \cdot 3 \end{bmatrix} + \begin{bmatrix} (-1) \cdot 5 \\ (-1) \cdot 1 \end{bmatrix} \quad \begin{matrix} \text{(Scalar multiplication)} \end{matrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} + \begin{bmatrix} -5 \\ -1 \end{bmatrix}$$